Data Mining: Classification

Fall 2026 (7761)

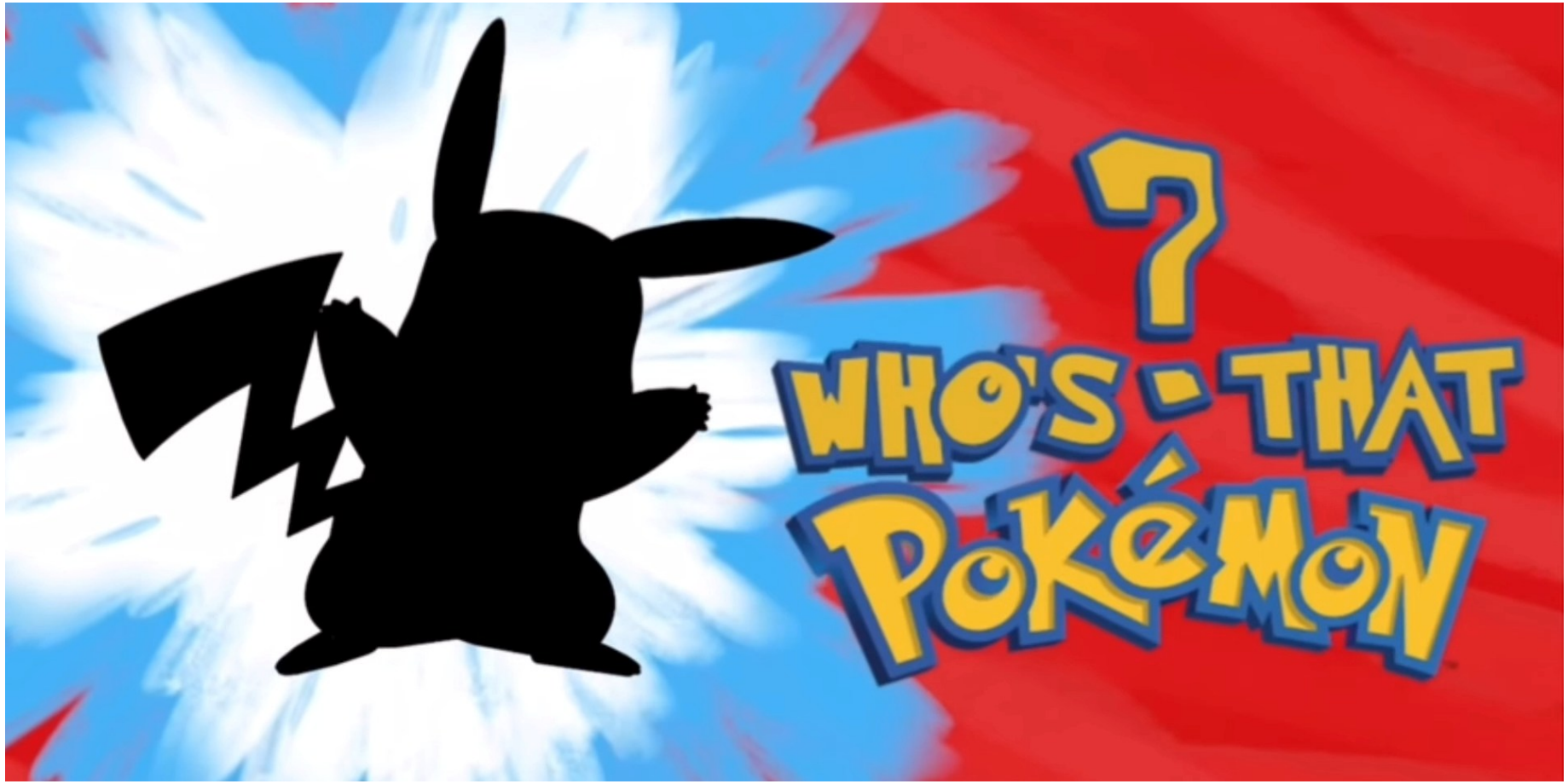
Dept. of Computer Science and Engineering

Instructor: Ting-Wei Lin

2026/10/23, Week 7

國立中興大學
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Can you identify the Pokémon?



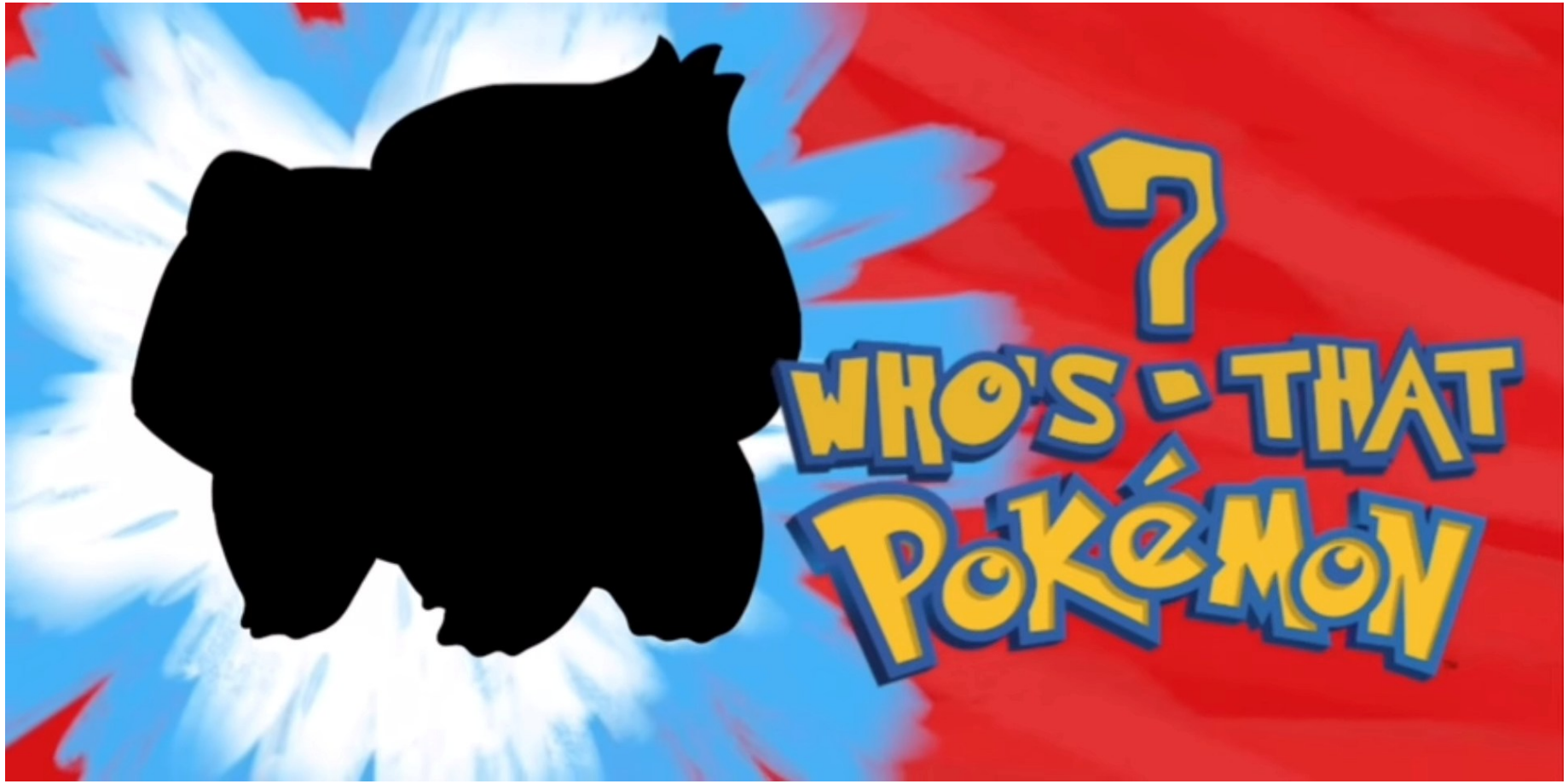
Input: an unlabeled image

Prediction



Predicted label: Pikachu

A new unlabeled example



Classification (cont.)



Predicted label: Bulbasaur or Pikachu

Classification in Data Mining

- Data mining is the process of extracting and finding patterns in massive data sets involving
  - Methods at the intersection of machine learning, statistics
- In data mining, we want
  - Discover interesting patterns, including models, and other useful knowledge from large datasets
- In classification, the learned classifier is an interesting pattern:
  - It captures the relationship between attributes and class labels.

The Class Label is also an Attribute

- Each object is described by multiple attributes.
  - One attribute is designated as the class label (target attribute).
- Example:
  - Image features: color, shape, texture, ...
  - Class label: Cat or Dog



Object 1



Object 2

In this example, each object is represented by an image.
Class label(a special target attribute): Animal type

- Object 1: Cat
- Object 2: Dog

What Is Classification?

- Classification learns from labeled examples
  - Each object has attributes (features) and a class label.
- Goal:
  - Given a new object with an unknown class label, predict which class it belongs to.
- Pokémon example
  - Attributes: color, ear shape, tail shape, body size, ...
  - Class label: Pikachu, Bulbasaur, Charmander, ...

Classification: A Formal View

- Given a collection of recordings (labeled training dataset)
  - $D_{train} = \{(x_i, y_i)\}_{i=1}^{n}$
    - $x_i \in X$: feature vector of object i
    - $y_i \in Y$: class label of object i
    - X: feature space
    - Y: set of possible class label
  - Classification learns a function
    - Learn a classifier $f: X \rightarrow Y, (y = f(x))$
  - For a new object x_z, the predicted class is
    - $\widehat{y_z} = f(x_z)$

How does classification work

- It has two main stages:

① Learning / Training (Supervised learning):

- Construct a classifier from labeled training data.
- Each training example is a pair (x_i, y_i)
  - $x_i = (x_{i1}, x_{i2}, \ldots, x_{id})$: d-dimensional feature vector
  - $y_i \in Y$: class label (discrete or nominal)
- The learning algorithm uses $D_{train} = \{(x_i, y_i)\}, i = 1, \ldots n$ to learn a classifier $f: X \rightarrow Y$
- $f_1(X) = \hat{Y}_1, f_2(X) = \hat{Y}_2, \ldots, f_i(X) = \hat{Y}_i$, which f_i is the best?
  - How can we choose a model and its hyperparameters without using the test set?
  - Use validation data

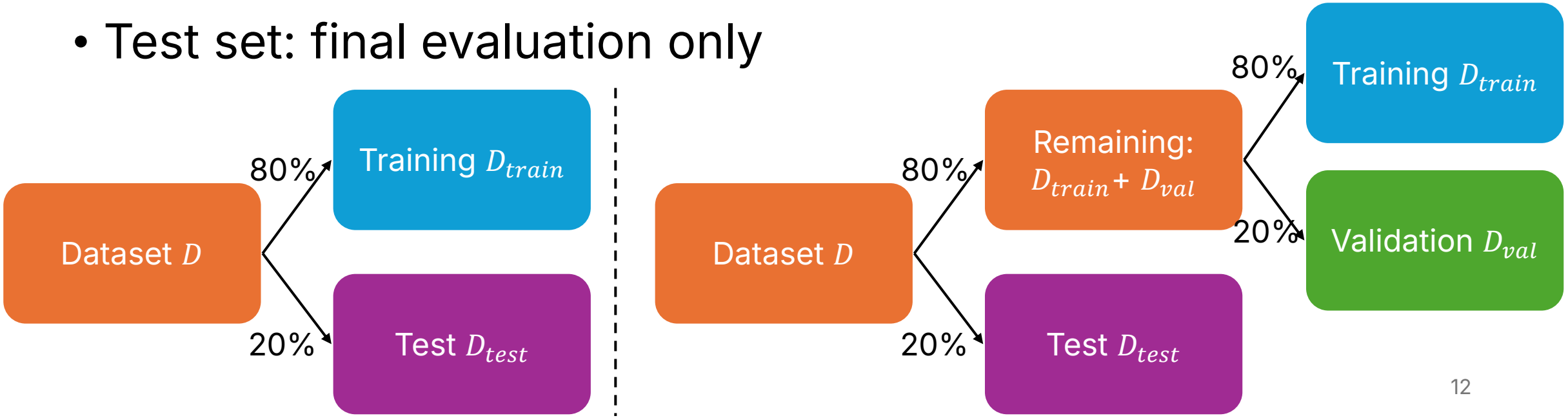
How does classification work

② Evaluation and Prediction:

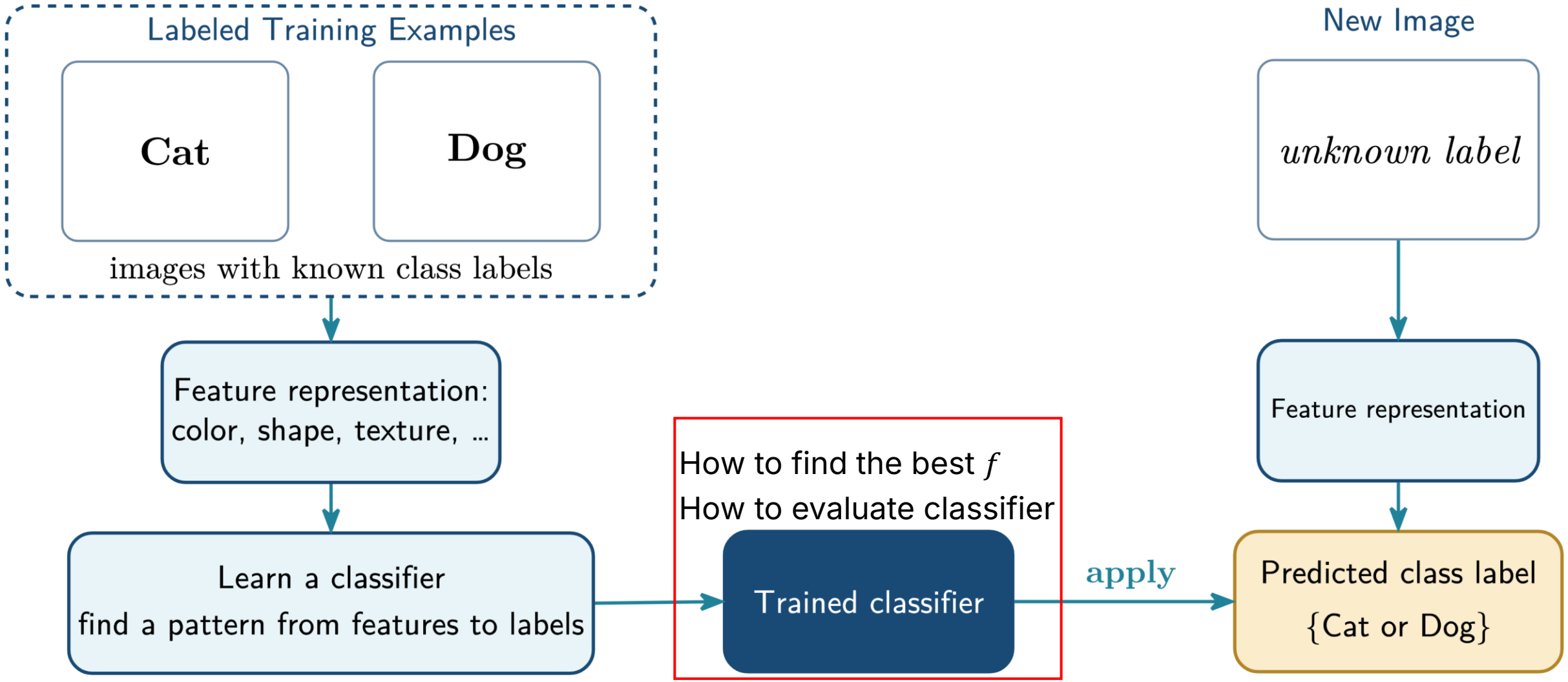
- Evaluate the classifier on held-out test data, then use it to predict labels for new unseen objects.
- Test examples are unseen by the learning and model-selection process, but their true labels are known for evaluation.
- Predict $\hat{y}_i = \hat{f}(x_i)$ for each test example
  - The classifier predicts it from the other attributes.
- Compare $\hat{y}_i$ with its known true label y_i
- If the result is acceptable, use the model to classify new data

Training, Validation, and Test Sets

- Training set: fit model parameters
- Validation set: choose model / tune hyperparameters
  - Examples: tree depth, k in k-NN, C and γ in SVM, or neural-network architecture.
- Test set: final evaluation only



Classification Pipeline



When engaging in classification tasks

1. Which features should we use?
   - Feature extraction and feature selection.
2. Which classification algorithm should we use?
   - Which hyperparameters should we choose?
   - For deep learning: architecture, activation functions, learning rate, batch size, ...
3. How should we split the labeled dataset?
   - Training set, validation set, and test set.
4. How should we evaluate model performance?

Classification Example: Instrument Recognition

- Input: an audio clip x
- Output (class label y): piano, guitar, violin, …
  1. Which features should we use?
     - MFCCs, spectral centroid, Whisper, …
  2. Which classification algorithm should we use?
     - SVM, Decision Tree, 2D Convolutional Neural Networks
  3. How should we split the labeled dataset?
     - 80% for training and 20% for test
  4. How should we evaluate model performance?
     - $Accuracy$=(# $correctly\ classified\ test\ sample$) / (# $test\ examples$)

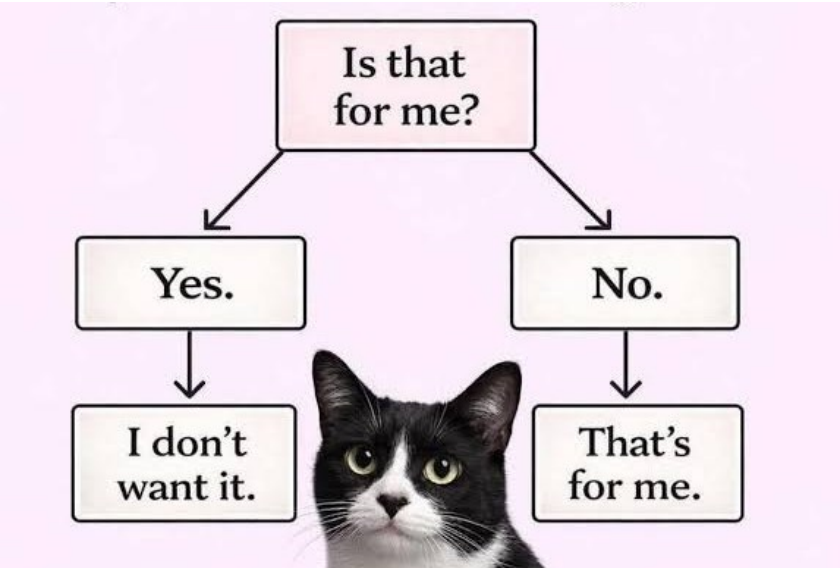
Classification Methods in This Course

- Decision Trees
- Naïve Bayes Classifier
- K-Nearest-Neighbors
- Case-Based Reasoning
- Logistic Regression
- Support Vector Machines
- Ensemble Classifiers

Decision Tree Induction

What Is a Decision Tree?

- A decision tree is a classifier represented as a flowchart-like tree.
  - Each internal node tests an attribute.
  - Each branch corresponds to an outcome of that test.
  - Each leaf node predicts a class label.





Source:
https://www.youtube.com/shorts/NGW1xdbOrm0

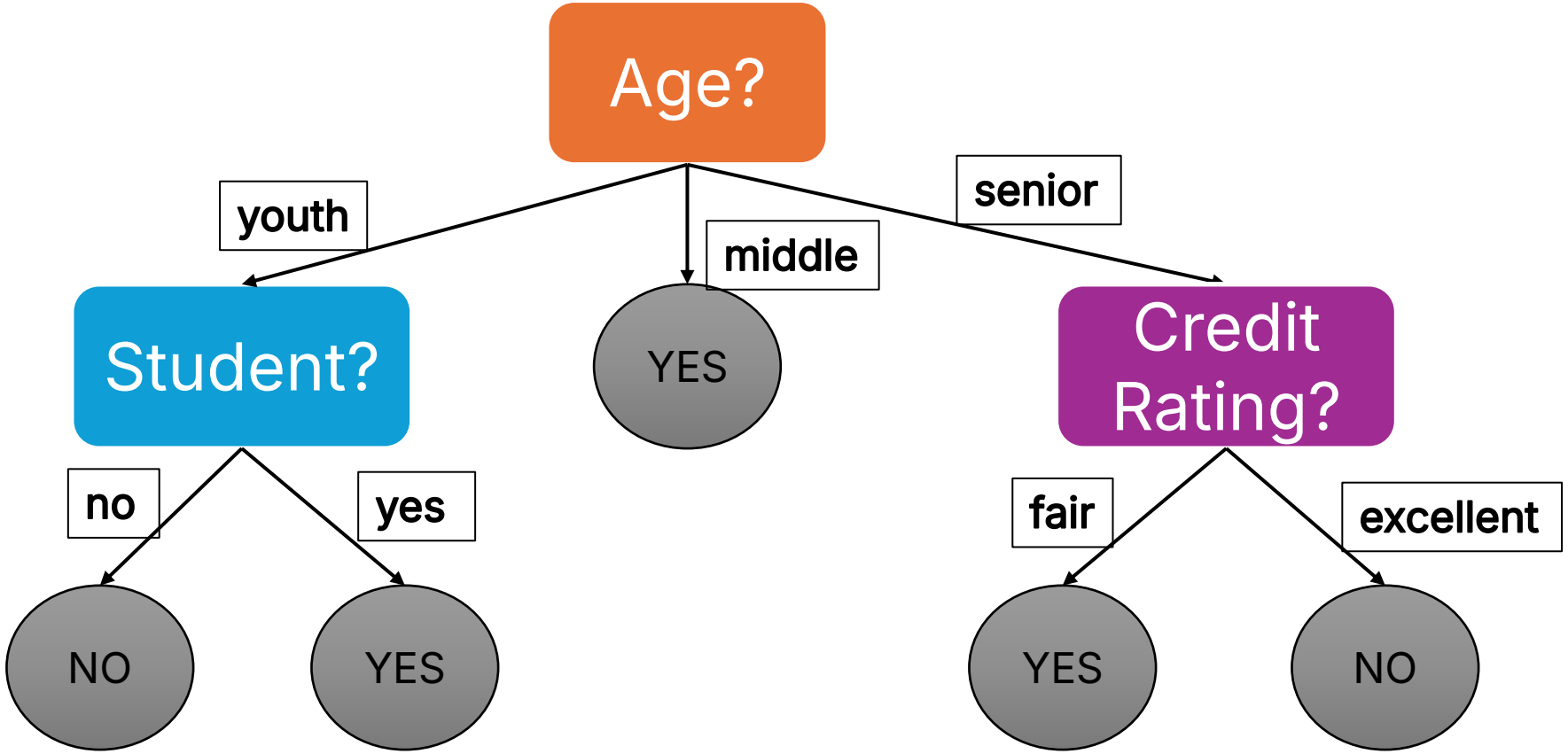
Classification by Decision Tree

- To classify a new instance, start at the root and follow the matching branches to a leaf.
  - At start, all the training examples are at the root
  - Examples are partitioned recursively based on selected attributes
  - On each node, attributes are selected based on the training examples on that node, and a heuristic or statistical measure

Training Data for Predicting buy computer

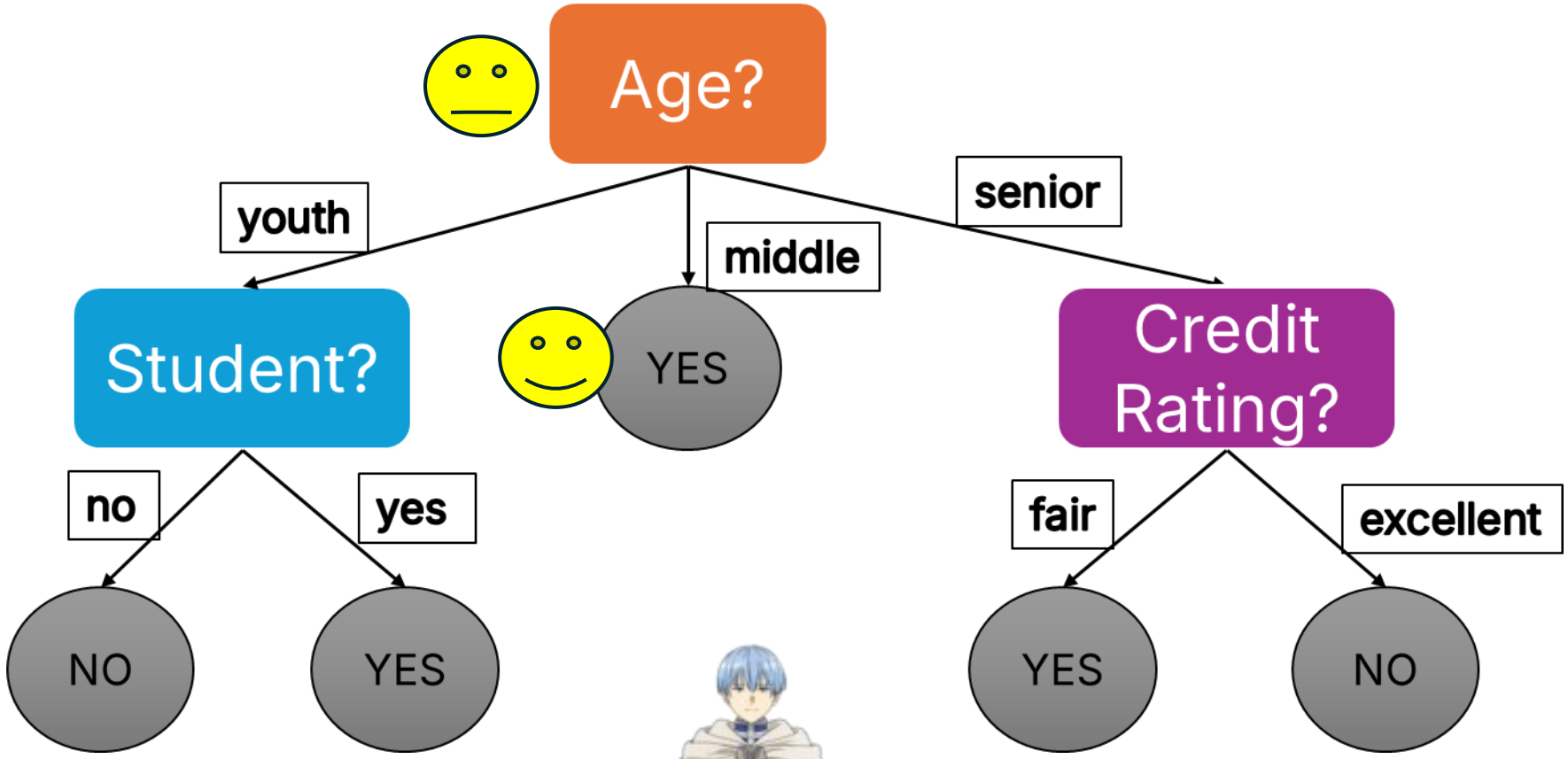
age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

Decision Tree Example



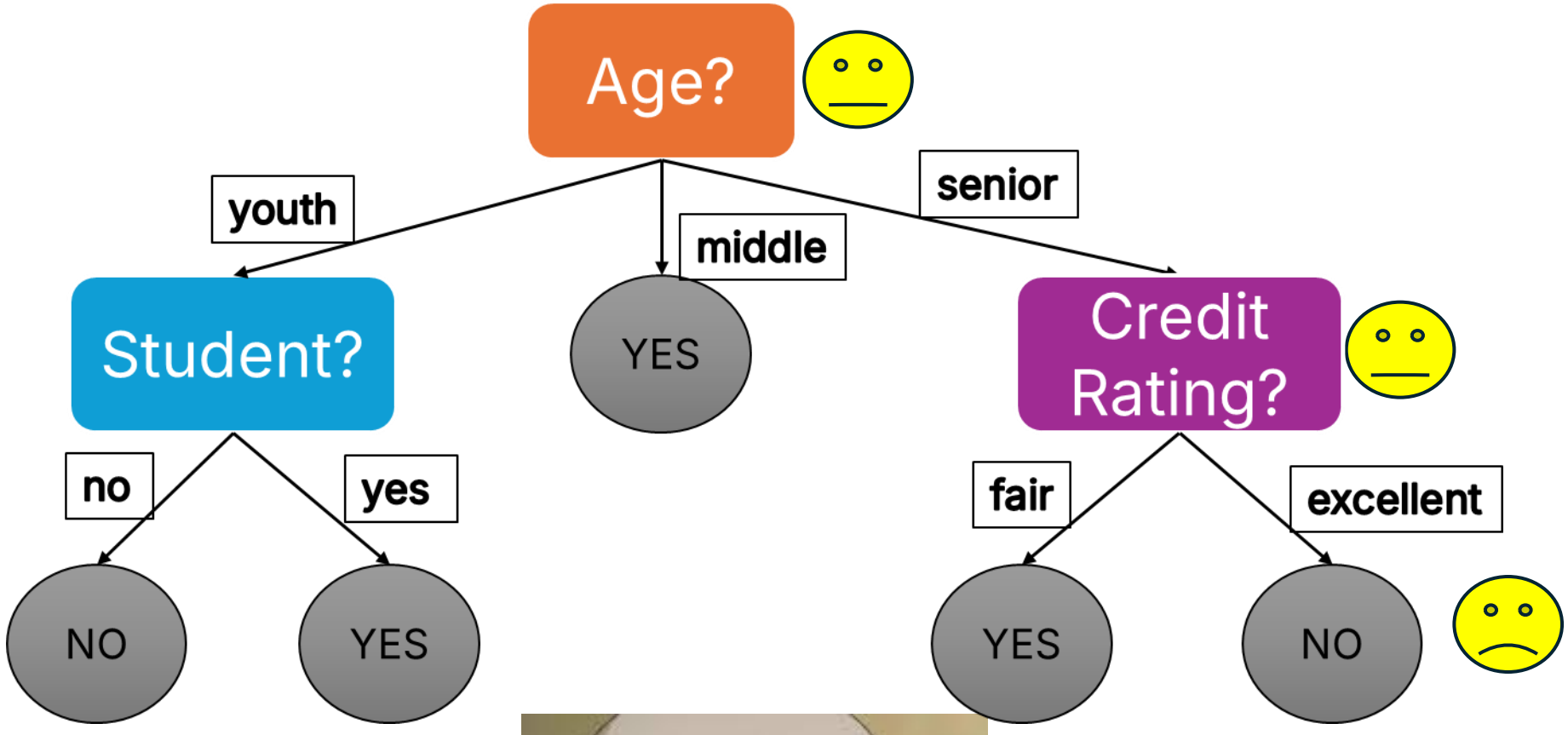
The tree predicts whether a customer will buy a computer. Each internal node tests an attribute and each leaf predicts yes or no.(either buy_computer = yes or buy_computer = no)


如果是勇者欣梅爾 的話，他會買電腦嗎?



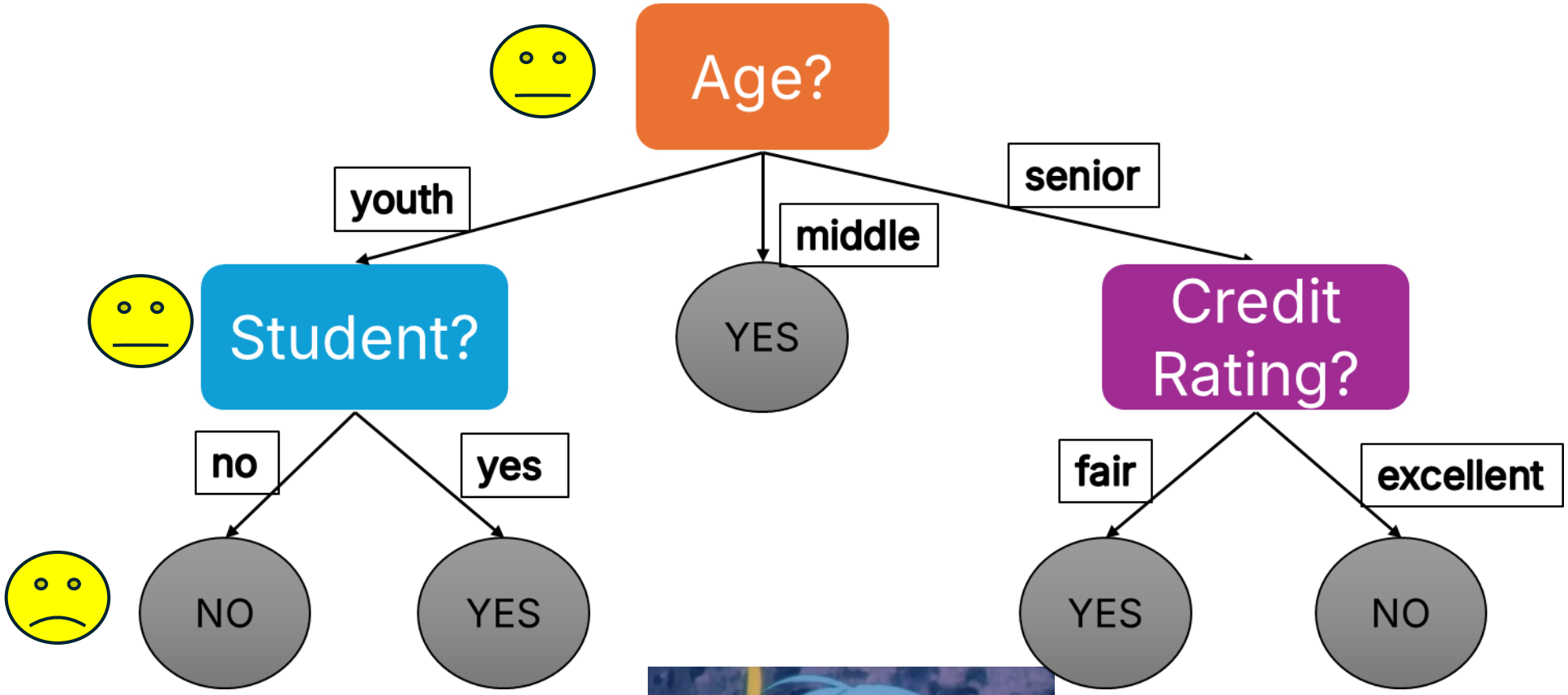
Customer
- Age=middle
- Income=high
- Student=no
- Credit rating=excellent





Customer
- Age=senior
- Income=medium
- Student=no
- Credit rating=excellent

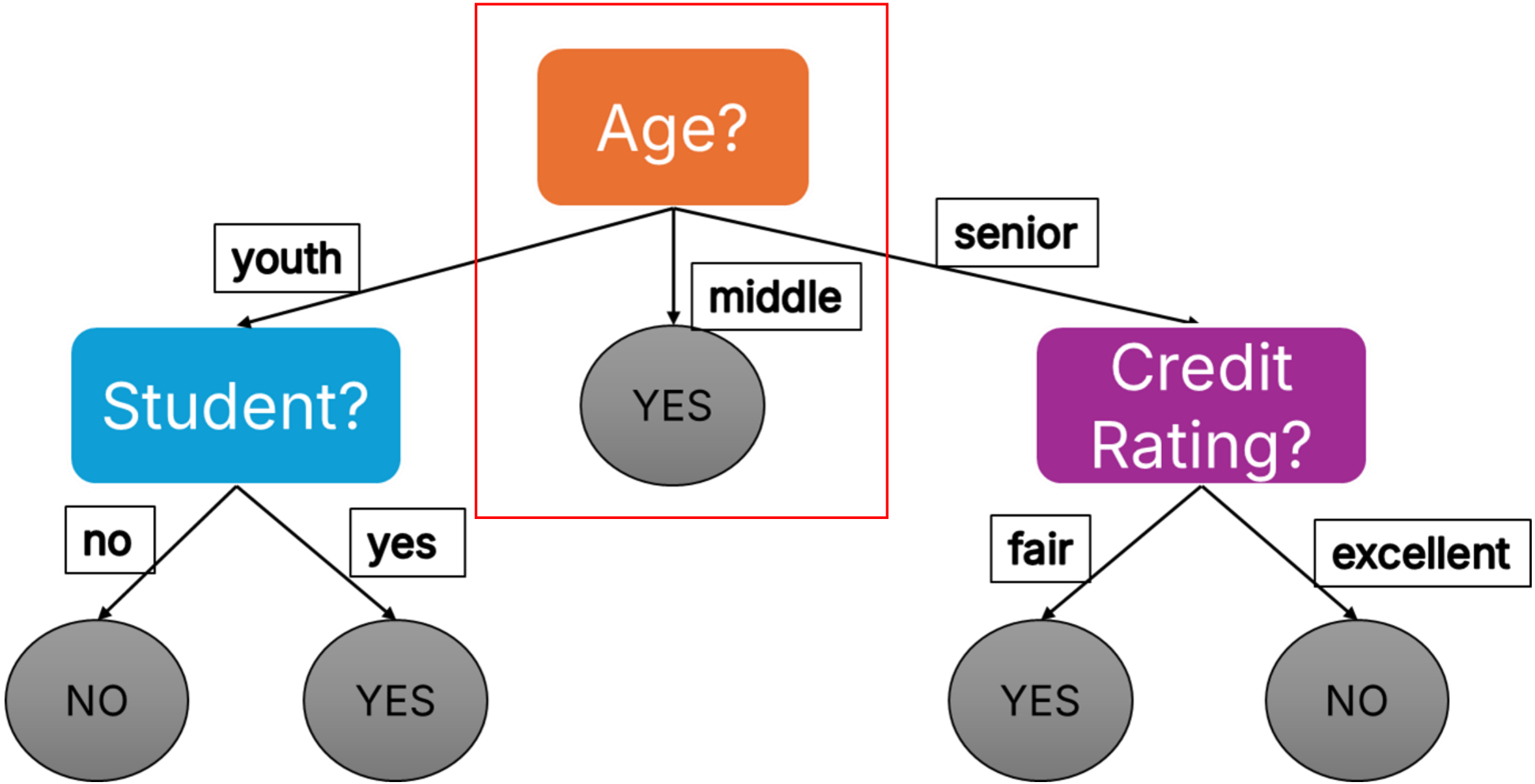




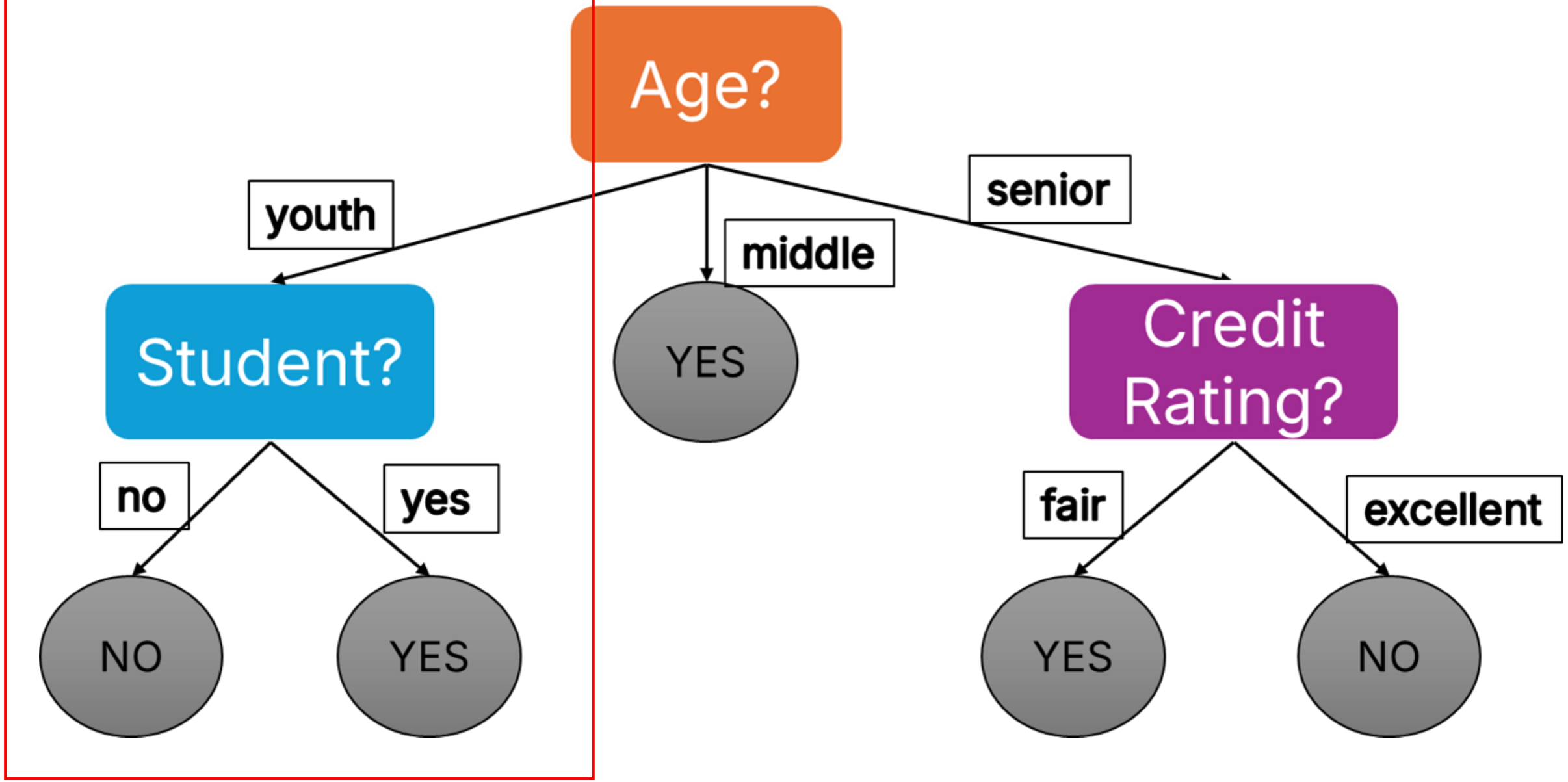
Customer
- Age=youth
- Income=low
- Student=no
- Credit rating=fair



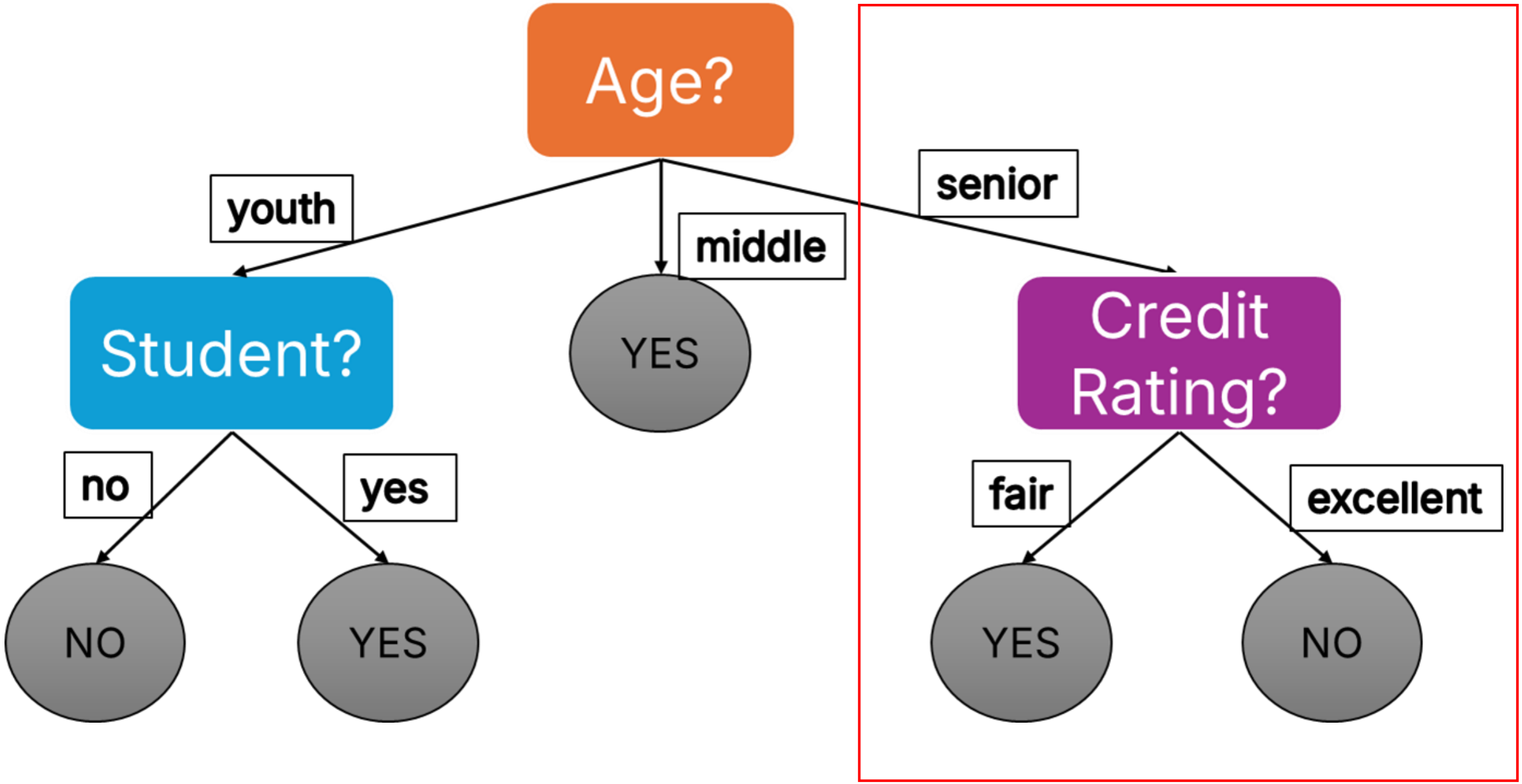
age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no


Age?
youth
middle
senior
Student?
YES
Credit Rating?
no
yes
fair
excellent
NO
YES
YES
NO

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no


Age?
youth
middle
senior
Student?
YES
Credit Rating?
no
yes
fair
excellent
NO
YES
YES
NO

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no


Age?
youth
middle
senior
Student?
YES
Credit Rating?
no
yes
fair
excellent
NO
YES
YES
NO

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

Income = medium
Buy: yes and no

Income = low
Buy: yes and no

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

Algorithm: Generate_decision_tree. Generate a decision tree from the training tuples of data partition, D.

Input:

- Data partition, D, which is a set of training tuples and their associated class labels;
- *attribute_list*, the set of candidate attributes;
- *Attribute_selection_method*, a procedure to determine the splitting criterion that "best" partitions the data tuples into individual classes. This criterion consists of a *splitting_attribute* and, possibly, either a *split-point* or *splitting subset*.

Output: A decision tree.
Method:

(1) create a node N;
(2) **if** tuples in D are all of the same class, C, **then**
(3) return N as a leaf node labeled with the class C;
(4) **if** *attribute_list* is empty **then**
(5) return N as a leaf node labeled with the majority class in D; // majority voting
(6) apply **Attribute_selection_method**(D, *attribute_list*) to **find** the "best" *splitting_criterion*;
(7) label node N with *splitting_criterion*;
(8) **if** *splitting_attribute* is discrete-valued **and**
multiway splits allowed **then** // not restricted to binary trees
(9) *attribute_list* $\leftarrow$ *attribute_list* $-$ *splitting_attribute*; // remove *splitting_attribute*
(10) **for each** outcome j of *splitting_criterion*
// partition the tuples and grow subtrees for each partition
(11) let D_j be the set of data tuples in D satisfying outcome j; // a partition
(12) **if** D_j is empty **then**
(13) attach a leaf labeled with the majority class in D to node N;
(14) **else** attach the node returned by **Generate_decision_tree**(D_j, *attribute_list*) to node N;
endfor
(15) return N.

Generic Decision Tree Algorithm

- Given a data partition D
  - ① Create a node N containing the examples in D.
  - ② If all examples in D have the same class, create N as a leaf.
  - ③ If no candidate attributes remain, create N as a majority-class leaf.
  - ④ Choose the best split according to an attribute-selection criterion.
  - ⑤ Partition D according to the split.
  - ⑥ Recursively grow one subtree for each partition.

After selecting age attribute

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

D_{youth}: 5

age	income	student	credit_rating	Class:buy_computer
middle	high	no	fair	yes
middle	low	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes

D_{middle}: 4

age	income	student	credit_rating	Class:buy_computer
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
senior	medium	yes	fair	yes
senior	medium	no	excellent	no

D_{senior}: 5

Pure node

② If all examples in D have the same class C, create N as a leaf labeled C.

age	income	student	credit_rating	Class:buy_computer
middle	high	no	fair	yes
middle	low	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes

D_{middle}

$D_{middle} = \{yes, yes, yes, yes\}$: Leaf → yes

No attributes remain

③ If no candidate attributes remain, return a leaf labeled with the majority class in D.

- Remaining candidate attributes: none

age	income	student	credit_rating	Class:buy_computer
middle	low	yes	excellent	yes
middle	low	yes	excellent	yes
middle	low	yes	excellent	yes
middle	low	yes	excellent	no

$\{yes, yes, yes, no\}$: majority-class leaf → yes

④ Choose the best split according to an attribute-selection criterion.

- Not Pure node and No attributes remain
- Attribute candidate: $\{income, student, credit_rating\}$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

Choose income

- $income = high, class \Rightarrow \{no, no\}$
- $income = medium, class \Rightarrow \{no, yes\}$
- $income = low, class \Rightarrow \{yes\}$

④ Choose the best split according to an attribute-selection criterion.

- Not Pure node and No attributes remain
- Attribute candidate: $\{income, student, credit_rating\}$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

Choose student

- $student = no, class \Rightarrow \{no, no, no\}$
- $student = yes, class \Rightarrow \{yes, yes\}$

④ Choose the best split according to an attribute-selection criterion.

- Not Pure node and No attributes remain
- Attribute candidate: $\{income, student, credit_rating\}$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

Choose credit_rating

- $credit_rating = fair, class \Rightarrow \{no, no, yes\}$
- $credit_rating = excellent, class \Rightarrow \{no, yes\}$

④ Choose the best split according to an attribute-selection criterion.

- Not Pure node and No attributes remain
- Attribute candidate: $\{income, student, credit_rating\}$

Choose income

- $income = high, class \Rightarrow \{no, no\}$
- $income = medium, class \Rightarrow \{no, yes\}$
- $income = low, class \Rightarrow \{yes\}$

Choose student

- $student = no, class \Rightarrow \{no, no, no\}$
- $student = yes, class \Rightarrow \{yes, yes\}$

Which one is better?

Choose credit_rating

- $credit_rating = fair, class \Rightarrow \{no, no, yes\}$
- $credit_rating = excellent, class \Rightarrow \{no, yes\}$

⑤ Partition D according to the split.

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
youth	medium	no	fair	no

age	income	student	credit_rating	Class:buy_computer
youth	low	yes	fair	yes
youth	medium	yes	excellent	yes

How Can an Attribute Split a Node?

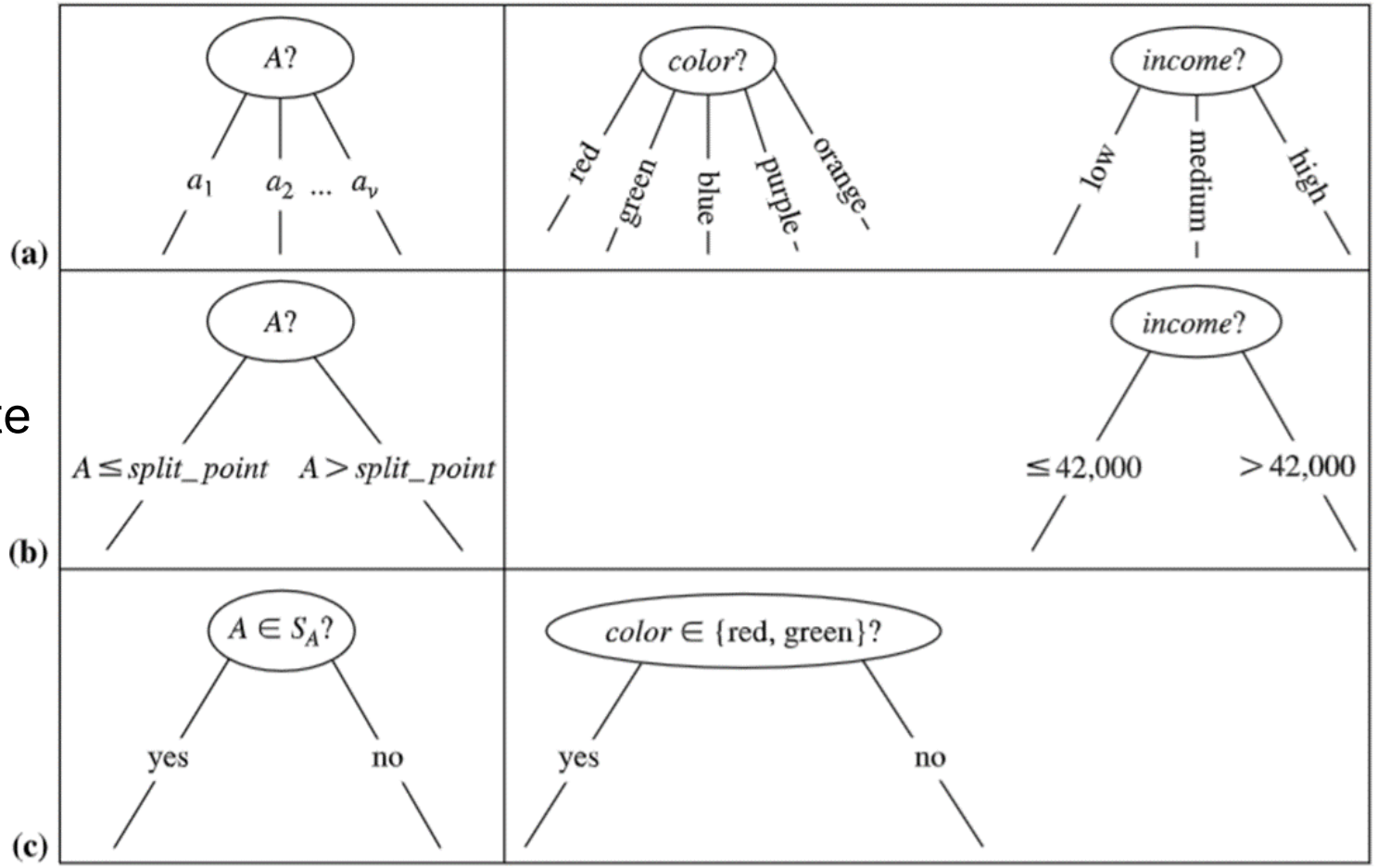
a) Categorical: multiway
   - One branch per value

b) Numerical
   - Threshold split

c) Categorical, binary
   - value-subset split

After splitting: update the candidate attribute list

- A discrete attribute is used at most once along any root-to-leaf path.
- Numerical attributes can be further used within different thresholds.



Handling Continuous Attributes

- Method 1: Discretization
  - Convert a continuous attribute into intervals
  - E.g., age: < 20, 20..30, 30..40, 40..50, > ➔ “20 - 50”
- Method 2: Learn a threshold at a node
  - Sort the observed values of attribute
    - E.g. 15, 18, 21, 22, 24, 25, 29, 31, …
  - Possible split point: $(a_i + a_{i+1})/2$

$$16.5, 19.5, 21.5, 23, 24.5, 27, 30, \ldots$$

Generic Decision Tree: Summary

① Start with all training examples at the root.

② Select the best split using an attribute-selection criterion.

③ Partition examples and recursively grow child nodes.

④ Stop when a node is pure, no useful attributes remain, or a branch receives no examples and create a leaf using the class or majority class.

Decision Trees: Strengths and Limitations

- Strengths
  - Interpretable: predictions can be expressed as if–then rules.
  - Require little preprocessing and feature scaling is usually unnecessary.
  - Handle categorical and numerical attributes.
  - Can work on various attribute types

Decision Trees: Strengths and Limitations

- Limitations
  - High variance: small changes in training data may produce a different tree.
  - Can overfit when the tree grows too deep.
  - Greedy splitting is locally optimal, not globally optimal.
  - A single tree may be less accurate than ensemble methods.

Attribute Selection Measures

How do we choose the best split at a node?

Information gain (ID3), Gain ratio (C4.5), Gini impurity (CART)

Decision Tree History

- Building on earlier work on concept learning systems (CLS), J. Ross Quinlan introduced ID3 in his 1986 paper “Induction of Decision Trees.”
- Quinlan later presented C4.5 (a successor of ID3)
  - C4.5 became an influential benchmark decision-tree algorithm and extended ID3 with gain ratio, continuous attributes, missing values, and pruning.



Ross Quinlan

FOLLOW

RuleQuest Research Pty Ltd
Verified email at rulequest.com

TITLE	CITED BY	YEAR
C4. 5: Programs for machine learning JR Quinlan Morgan Kaufmann, San Francisco, CA	46453 *	1993
Induction of decision trees JR Quinlan Machine learning 1 (1), 81-106	35092	1986

Decision Tree History (cont.)

- In 1984, a group of statisticians (L. Breiman, J. Friedman, R. Olshen, and C. Stone) published the book Classification and Regression Trees (CART), which described the generation of binary decision trees
- ID3, C4.5, and CART
  - Which split produces the purest child nodes? (a pure node contains examples from only one class.)

Information gain (ID3)

How purity is a node?

Why Do We Need a Measure of Purity?

student	buy_computer
no	no
no	no
no	no
no	no
yes	yes
yes	yes
yes	yes
no	no
yes	yes
yes	yes
no	no
yes	yes

s\c	Y	N
Y	6/6	0/6
N	0/6	6/6

A good split makes its child nodes as pure as possible.

student	buy_computer
no	yes
no	no
no	yes
no	no
yes	yes
yes	no
yes	no
no	yes
yes	yes
yes	no
no	no
yes	yes

s\c	Y	N
Y	3/6	3/6
N	3/6	3/6

Entropy: A Measure of Class Uncertainty

- Proposed by Claude E. Shannon of Bell Labs

A mathematical theory of communication

CE Shannon - The Bell system technical journal, 1948 - ieeexplore.ieee.org

… By a communication system we will mean a system of the type … as mathematical entities, suitably idealized from their physical counterparts. We may roughly classify communication …

☆ 儲存 99 引用 被引用 118531 次 相關文章 全部共 85 個版本

Quantities of the form $H = -\sum p_i \log p_i$ (the constant K merely amounts to a choice of a unit of measure) play a central role in information theory as measures of information, choice and uncertainty. The form of H will be recognized as that of entropy as defined in certain formulations of statistical mechanics[8] where p_i is the probability of a system being in cell i of its phase space. H is then, for example, the H in Boltzmann's famous H theorem. We shall call $H = -\sum p_i \log p_i$ the entropy of the set of probabilities $p_1, \ldots, p_n$. If x is a chance variable we will write $H(x)$ for its entropy; thus x is not an argument of a function but a label for a number, to differentiate it from $H(y)$ say, the entropy of the chance variable y.

The entropy in the case of two possibilities with probabilities p and $q = 1 - p$, namely

$$H = -(p \log p + q \log q)$$

Entropy is 0 when a node is pure, and largest when class labels are evenly mixed.

Entropy (information theory)

- Entropy of message M with alphabet $\{m_1, m_2, \dots, m_n\}$

$$H(M) = -\sum_{i=1}^{n} p(m_i) \log_2 p(m_i)$$

  - $p(m_i)$ is the probability that m_i in M will occur
- Example: $\mathrm{M}_1 = \{m_1, m_2\}, \mathrm{M}_2 = \{m_1, m_1, m_2\}, \mathrm{M}_3 = \{m_1, m_1, m_1\}$

1. $p(m_1) = \frac{1}{2}, p(m_2) = \frac{1}{2} : H(M) = -\left(\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right) = -(-0.5 + -0.5) = 1$
2. $p(m_1) = \frac{2}{3}, p(m_2) = \frac{1}{3} : H(M) = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) = -(-0.39 + -0.53) = 0.92$
3. $p(m_1) = \frac{3}{3}, p(m_2) = \frac{0}{3} : H(M) = -(\frac{3}{3}\log_2\frac{3}{3} + \frac{0}{3}\log_2\frac{0}{3}) = -(0 + 0) = 0$

Entropy is highest when classes are evenly distributed, and lowest when one class dominates.

Information gain (ID3)

- ID3 uses information gain (based on entropy) as its attribute selection measure
- Suppose we were to partition the tuples in D on some attribute A (discrete-valued) having v distinct values, $\{a_1, a_2, \ldots, a_v\}$, with class set $C = \{c_1, c_2, \ldots, c_k\}$.
- Split D into v partitions or subsets, $\{D_1, D_2, \ldots, D_v\}$,
  - $Info(D) = -\sum_{i=1}^{k} p(c_i) \log_2 p(c_i)$
  - $Info_A(D) = \sum_{j=1}^{v} \frac{D_j}{D} Info(D_j)$
    - Weighted average entropy after splitting D on attribute A.
  - $Gain(A) = Info(D) - Info_A(D)$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

age	Class
senior	yes
senior	yes
senior	no
senior	yes
senior	no
senior	no
middle	yes
youth	no
youth	yes
senior	yes
youth	yes
middle	yes
middle	yes
senior	no

$$Info(D) = -\sum_{i=1}^{k} p(c_i)\log_2 p(c_i) \qquad Info_A(D) = \sum_{j=1}^{v} \frac{D_j}{D} Info(D_j)$$

Class
- Yes: 9
- No: 5

$$Info(D) = -(\frac{9}{9+5}\log_2\frac{9}{9+5} + \frac{5}{9+5}\log_2\frac{5}{9+5}) = 0.94$$

Partition 14 samples in attribute age
- a_1: youth, D_1: 5
- a_2: middle, D_2: 4
- a_3: senior, D_3: 5

Class
- Yes: 2
- No: 0

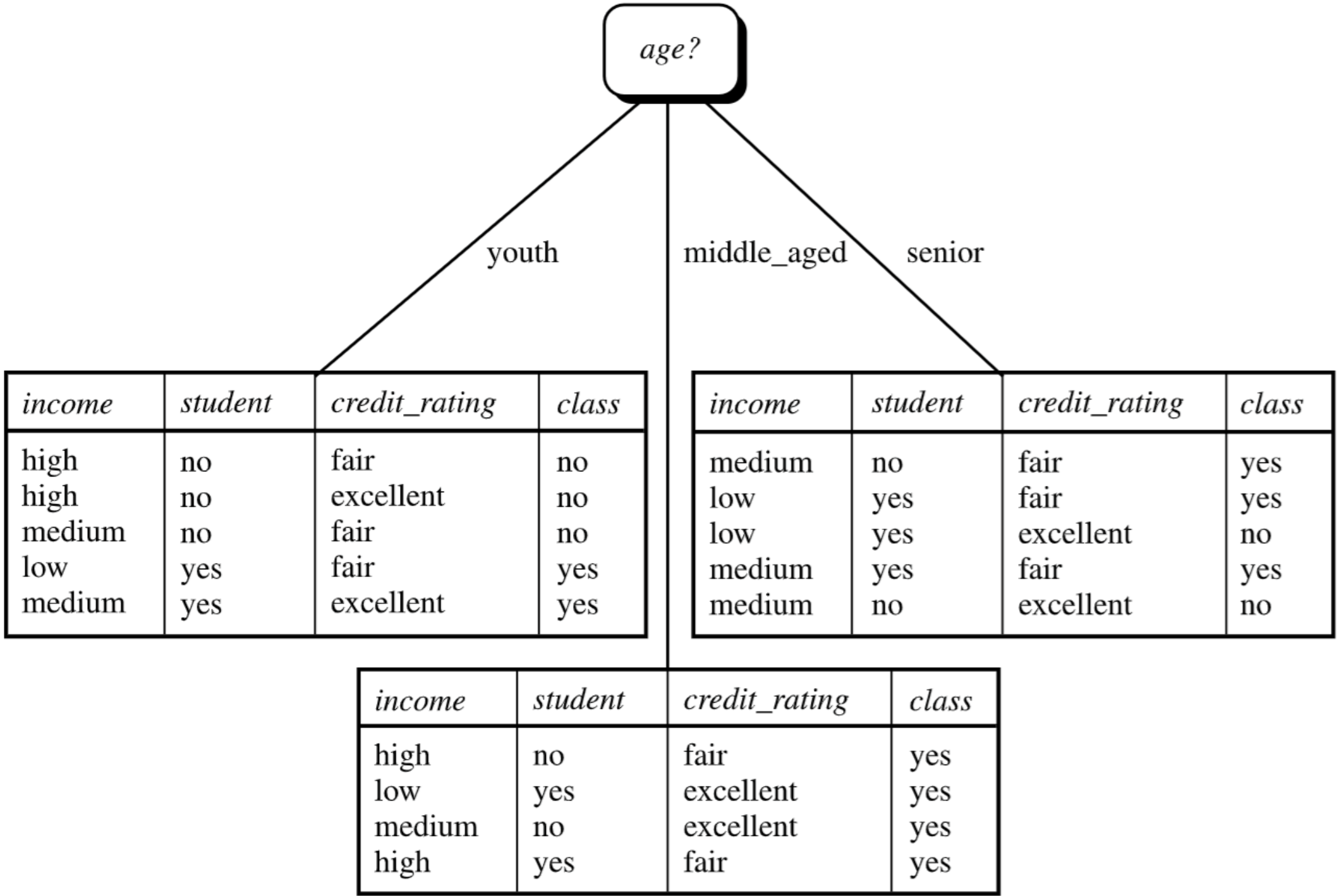
$$Info_{age}(D) = \frac{5}{5+4+5}(-(\frac{2}{3+2}\log_2\frac{2}{3+2} + \frac{3}{3+2}\log_2\frac{3}{3+2})) +$$
$$\frac{4}{5+4+5}(-(\frac{4}{4+0}\log_2\frac{4}{4+0} + \frac{0}{4+0}\log_2\frac{0}{4+0})) +$$
$$\frac{5}{5+4+5}(-(\frac{3}{3+2}\log_2\frac{3}{3+2} + \frac{2}{3+2}\log_2\frac{2}{3+2}))$$

$$= 0.694$$

$$Gain(age) = Info(D) - Info_{age}(D) = 0.94 - 0.694 = 0.246$$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

- $Gain(age) = 0.246$
- $Gain(income) = 0.029$
- $Gain(student) = 0.151$
- $Gain(credit_rating) = 0.048$



income	*student*	*credit_rating*	*class*
high	no	fair	no
high	no	excellent	no
medium	no	fair	no
low	yes	fair	yes
medium	yes	excellent	yes

income	*student*	*credit_rating*	*class*
medium	no	fair	yes
low	yes	fair	yes
low	yes	excellent	no
medium	yes	fair	yes
medium	no	excellent	no

income	*student*	*credit_rating*	*class*
high	no	fair	yes
low	yes	excellent	yes
medium	no	excellent	yes
high	yes	fair	yes

Gain ratio (C4.5)

If we have too much branch?

Gain ratio (C4.5)

- The limitation of information gain:
  - The information gain measure is biased toward splits with many branches / outcomes
  - Consider an attribute as a unique identifier split on it would result in a large number of partitions, each one containing just one tuple (entropy = 0)
- Gain ratio (normalization to information gain)
  - Using a "split information" value (number of count in each branch)

$$SplitInfo_A(D) = -\sum_{j=1}^{v} \frac{|D_j|}{|D|} \log_2 \frac{|D_j|}{|D|}$$

$$GainRatio(A) = \frac{Gain(A)}{SplitInfo_A(D)}$$

age	Class
youth	no
youth	no
middle	yes
senior	yes
senior	yes
senior	no
middle	yes
youth	no
youth	yes
senior	yes
youth	yes
middle	yes
middle	yes
senior	no

$$Info(D) = -(\frac{9}{9+5}\log_2\frac{9}{9+5} + \frac{5}{9+5}\log_2\frac{5}{9+5}) = 0.94$$

$$Info_{age}(D) = \frac{5}{5+4+5}(-(\frac{2}{3+2}\log_2\frac{2}{3+2} + \frac{3}{3+2}\log_2\frac{3}{3+2})) +$$

$$\frac{4}{5+4+5}(-(\frac{4}{4+0}\log_2\frac{4}{4+0} + \frac{0}{4+0}\log_2\frac{0}{4+0})) +$$

$$\frac{5}{5+4+5}(-(\frac{3}{3+2}\log_2\frac{3}{3+2} + \frac{2}{3+2}\log_2\frac{2}{3+2}))$$

$$= 0.694$$

Age
- youth: 5
- middle: 4
- Senior: 5

$$Gain(age) = Info(D) - Info_{age}(D) = 0.94 - 0.694 = 0.246$$

$$SplitInfo_{age} = -(\frac{5}{5+4+5}\log_2\frac{5}{5+4+5} + \frac{4}{14}\log_2\frac{4}{14} + \frac{5}{14}\log_2\frac{5}{14}) = 1.577$$

$$GainRatio(age) = \frac{0.246}{1.577} = 0.156$$

age	income	student	credit_rating	Class:buy_computer
youth	high	no	fair	no
youth	high	no	excellent	no
middle	high	no	fair	yes
senior	medium	no	fair	yes
senior	low	yes	fair	yes
senior	low	yes	excellent	no
middle	low	yes	excellent	yes
youth	medium	no	fair	no
youth	low	yes	fair	yes
senior	medium	yes	fair	yes
youth	medium	yes	excellent	yes
middle	medium	no	excellent	yes
middle	high	yes	fair	yes
senior	medium	no	excellent	no

- $Gain(age) = 0.246$
- $GainRatio(age) = 0.156$
- $Gain(income) = 0.029$
- $GainRatio(income) = 0.018$
- $Gain(student) = 0.151$
- $GainRatio(student) = 0.151$
- $Gain(credit_rating) = 0.048$
- $GainRatio(credit_rating) = 0.048$

$$SplitInfo_{age}(D) = -\frac{5}{14}\log_2\left(\frac{5}{14}\right) - \frac{4}{14}\log_2\left(\frac{4}{14}\right) - \frac{5}{14}\log_2\left(\frac{5}{14}\right) = 1.577$$

Gini impurity (CART)

How impure a dataset's class labels are within a node

Gini impurity

- Gini Impurity measures how "mixed" a partition is
  - The probability of misclassification when a class label is randomly assigned according to the node's class distribution.

$$\text{Gini} = \sum_{i=1}^{k} p_i(1 - p_i) = \sum_{i=1}^{k} (p_i - {p_i}^2) = \sum_{i=1}^{k} (p_i) - \sum_{i=1}^{k} ({p_i}^2) = 1 - \sum_{i=1}^{k} ({p_i}^2)$$

  - Suppose we have 3 class (A, B, C)
    - If a partition is pure (all A)

$$Gini = 1 * 0 + 0 * 1 + 0 * 1 = 0$$

    - If a partition has 30% of A, 40% of B, and 30% of C

$$Gini = 0.3 * 0.7 + 0.4 * 0.6 + 0.3 * 0.7 = 0.66$$

Gini impurity (CART)

- If a data set D contains examples from k classes, Gini index,
  - $Gini(D)$ is defined as

$$Gini(D) = 1 - \sum_{i=1}^{k} {p_i}^2$$

  - p_i is the proportion of examples in D that belongs to class C_i ($p_i = \frac{|D_{C_i}|}{|D|}$)

Class
- Yes: 9
- No: 5

$$Gini(D) = 1 - \left(\left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2\right) = 0.459$$

Gini impurity in CART

- CART uses Gini impurity and restricts each split to two branches
- When considering a binary split, we compute a weighted sum of the impurity of each resulting partition

$$\mathrm{G}ini_{split}(D) = \frac{|D_L|}{|D|} Gini(D_L) + \frac{|D_R|}{|D|} Gini(D_R)$$

- The reduction in impurity

$$\triangle Gini(split) = Gini(D) - \mathrm{G}ini_{split}(D)$$

age	Class
youth	no
youth	no
middle	yes
senior	yes
senior	yes
senior	no
middle	yes
youth	no
youth	yes
senior	yes
youth	yes
middle	yes
middle	yes
senior	no

$$Gini(D) = 1 - \sum_{i=1}^{k} {p_i}^2$$

Age

- youth vs {middle, senior}

$$Gini_{youth} = \frac{5}{14}\left(1 - \left(\left(\frac{2}{5}\right)^2 + \left(\frac{3}{5}\right)^2\right)\right) + \frac{9}{14}\left(1 - \left(\left(\frac{7}{9}\right)^2 + \left(\frac{2}{9}\right)^2\right)\right) = 0.357$$

- middle vs {youth, senior}

$$Gini_{middle} = \frac{4}{14}\left(1 - \left(\frac{4}{4}\right)^2\right) + \frac{10}{14}\left(1 - \left(\left(\frac{5}{10}\right)^2 + \left(\frac{5}{10}\right)^2\right)\right) = 0.394$$

- senior vs {youth, middle}

$$Gini_{senior} = \frac{5}{14}\left(1 - \left(\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2\right)\right) + \frac{9}{14}\left(1 - \left(\left(\frac{6}{9}\right)^2 + \left(\frac{3}{9}\right)^2\right)\right) = 0.457$$

Gini impurity considers a binary split for each attribute: $Gini_{split}(D) = \frac{|D_L|}{|D|}Gini(D_L) + \frac{|D_R|}{|D|}Gini(D_R)$

Result of Gini

$$\triangle Gini(split) = Gini(D) - Gini_{split}(D)$$

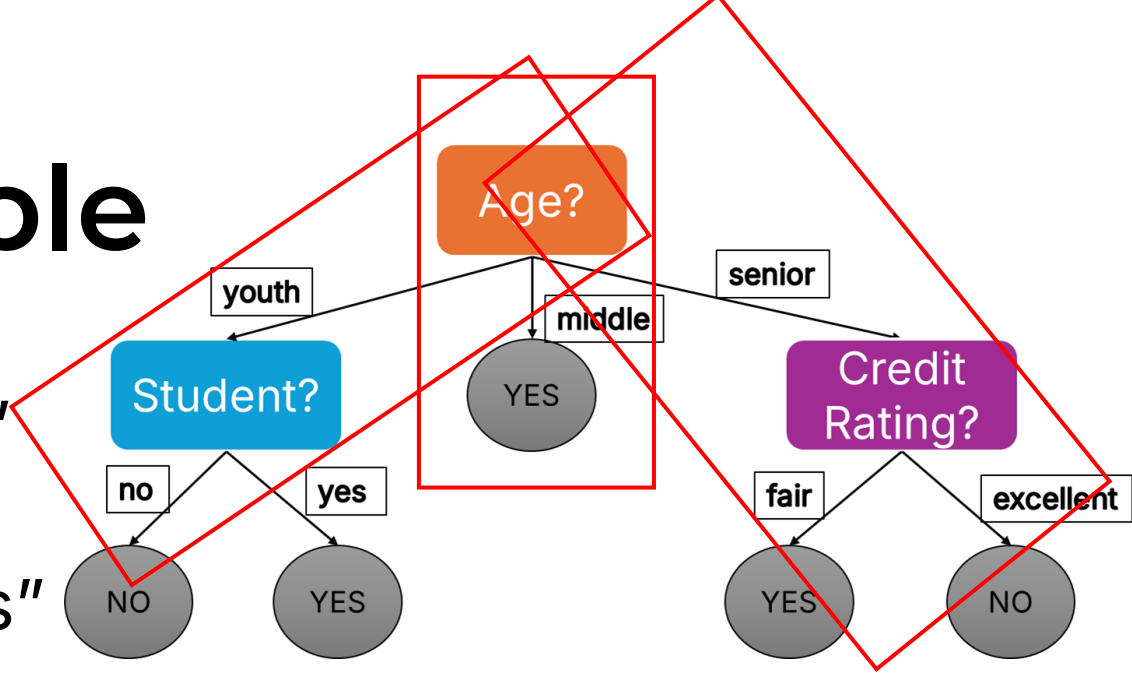
Attribute	Binary split	Gini
age	youth vs {middle, senior}	0.394
	middle vs {youth, senior}	0.357
	senior vs {youth, middle}	0.457
income	low vs {medium, high}	0.450
	medium vs {low, high}	0.458
	high vs {low, medium}	0.443
student	yes vs no	0.367
credit_rating	excellent vs fair	0.429

Can the attribute age be removed from the attribute candidate list?

From a Decision Tree to IF–THEN Rules

- Represent the knowledge in the form of IF-THEN rules
  - One rule is created for each path from the root to a leaf
  - Each split condition along a path forms a conjunction.
  - The leaf node provides the predicted class label.
- Rules are easier for humans to understand

Decision Tree Example



- IF age = “youth” AND student = “no”
  - THEN buy_computer = “no”
- IF age = “youth” AND student = “yes”
  - THEN buy_computer = “yes”
- IF age = “middle”
  - THEN buy_computer = “yes”
- IF age = “senior” AND credit_rating = “excellent”
  - THEN buy_computer = “no”
- IF age = “senior” AND credit_rating = “fair”
  - THEN buy_computer = “yes”

Optional: Regression tree

Table 6.2 Training data for regression.

attribute x_i	1	2	3	4	5
output y_i	10	12	8	20	22

Split point: 1.5
Left node value: 10
Right node value: 15.5 ((12+8+20+22)/4)

Split point: 2.5
Left node value: 11 ((10+12)/2)
Right node value: 16.7 ((8+20+22)/3)

candidate split point x_i	1.5	2.5	3.5	4.5
predicted value of left leaf node y_l	10	11	10	12.5
predicted value of right leaf node y_r	15.5	16.7	21	22
RSS	131	116.67	10	83

Choose the split with the smallest RSS.

$$\text{RSS}(residual\ sum\ of\ squares) = \sum_i (y_i - \hat{y}_i)^2$$

In regression trees, a good split makes the target values within each child node as homogeneous as possible.

Bayes Classification Methods

What are Bayesian classifiers

- Bayesian classifiers are statistical classifiers
  - Estimate the probability that a given tuple X belongs to each class Y
  - Based on Bayes' Theorem
- We will introduce a simple Bayesian classifier: the naïve Bayes classifier

Bayes' Theorem: Basics

- Bayes' Theorem:

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

- Let H be a hypothesis
  - E.g., H: the customer buys a computer.
- Let $X = (x_1, \ldots, x_d)$ denote the observed attributes vector of that customer.

Bayes' Theorem: Basics (cont.)

- Bayes' Theorem:

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

posterior probability

  - Suppose that H is the hypothesis that our customer will buy a computer and X is the customer's age and income.
  - $P(H|X)$: the probability that the customer buys a computer, given the observed attributes X

Bayes' Theorem: Basics (cont.)

- Bayes' Theorem:

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

prior probability

  - $P(H)$: the probability that a customer buys a computer before observing the attributes X

Bayes' Theorem: Basics (cont.)

- Bayes' Theorem:

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

Likelihood

  - $P(X|H)$: how likely the observed attributes X are if the customer buys a computer.

Bayes' Theorem: Basics (cont.)

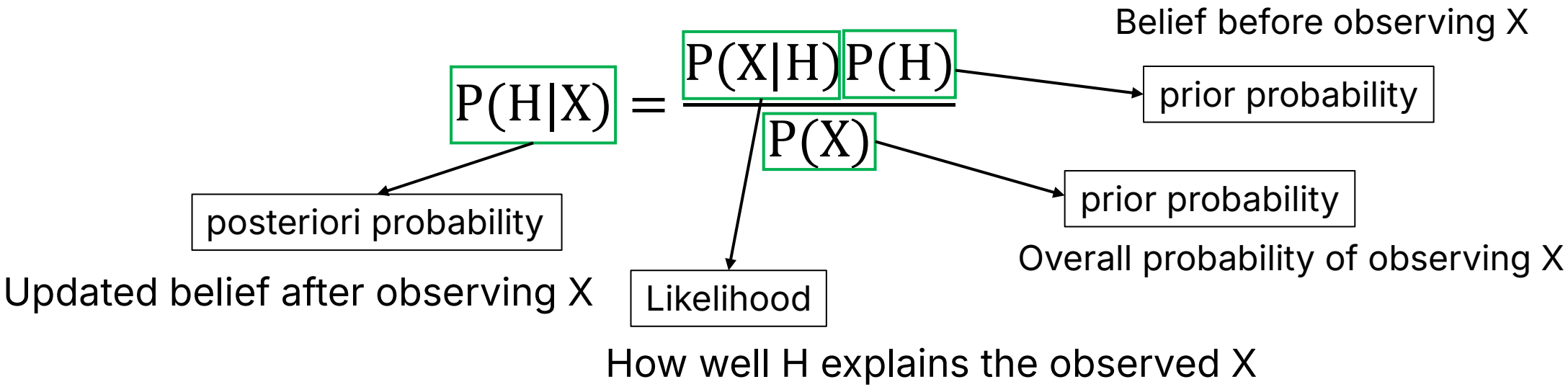
- Bayes' Theorem:

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

Evidence

  - $P(X)$: the overall probability of observing the attribute vector X, without conditioning on H

Bayes' Theorem: Basics (cont.)



Bayes' Theorem: Example

$$P(H|X) = \frac{P(X|H)P(H)}{P(X)}$$

- The morning is cloudy
  - What is the chance of rain?

$$P(rain|cloudy)$$

  - 50% of all rainy days start off cloudy

$$P(cloudy|rain) = 50\%$$

  - This is usually a dry month (only 3 of 30 days rainy)

$$P(rain) = 10\%$$

  - Cloudy mornings are common (40% of days start cloudy)

$$P(cloudy) = 40\%$$

  - Posterior probabilities

$$P(rain|cloudy) = \frac{P(cloudy\,|\,rain)P(rain)}{P(cloud)} = \frac{50\% \times 10\%}{40\%} = 0.125$$

Bayes' Theorem: Example (cont.)

- A Bayesian classifier computes the posterior probability for each of the classes and predicts the class with the highest posterior probability.
  - $P(buy\ computer|X)\ and\ P(not\ buy\ computer|X)$
  - $\hat{C} = argmax_{c_i \in C} P(C_i|X) = argmax_{c_i \in C} P(X|C_i)P(C_i)$
  - For a fixed observed tuple X, $P(X)$ is the same for all candidate classes. It does not affect the final class decision.

Naïve Bayes Classification

- Computing the joint likelihood $P(X|C_i)$, can be difficult when X contains many attributes.
  - $P(X|C_i) = P(x_1, x_2, \dots, x_d|C_i) = P(x_1|C_i)P(x_2|x_1, C_i) \dots P(x_d|x_{d-1}, \dots, x_1, C_i)$
  - Joint probability over many variables
- Naïve Bayes assumption : the attributes are conditionally independent given the class C_i
  - $P(X|C_i) = \prod_{k=1}^{d} P(x_k|\mathrm{C}_i) = P(x_1|\mathrm{C}_i)P(x_2|\mathrm{C}_i)P(x_3|\mathrm{C}_i) \dots P(x_d|\mathrm{C}_i)$
  - $P(x_1|\mathrm{C}_i), \dots, P(x_d|\mathrm{C}_i)$ can be easily estimated from the training tuples
    - Estimate the conditional probability for each feature x_1 and each class C_i

Outlook	Temperature	Humidity	Windy	Play golf
rainy	hot	high	false	no
rainy	hot	high	true	no
overcast	hot	high	false	yes
sunny	mild	high	false	yes
sunny	cool	normal	false	yes
sunny	cool	normal	true	no
overcast	cool	normal	true	yes
rainy	mild	high	false	no
rainy	cool	normal	false	yes
sunny	mild	normal	false	yes
rainy	mild	normal	true	yes
overcast	mild	high	true	yes
overcast	hot	normal	false	yes
sunny	mild	high	true	no

Class:

- play golf=yes: 9
- play golf=no: 5

$$\prod_{k=1}^{d} P(x_k|\mathrm{C}_i)$$

Outlook	Play golf
rainy	no
rainy	no
overcast	yes
sunny	yes
sunny	yes
sunny	no
overcast	yes
rainy	no
rainy	yes
sunny	yes
rainy	yes
overcast	yes
overcast	yes
sunny	no

Class:

- play golf=yes: 9
- play golf=no: 5

sunny
play golf=yes: 3
play golf=no: 2

overcast
play golf=yes: 4
play golf=no: 0

rainy
play golf=yes: 2
play golf=no: 3

Frequency table	Yes	No
sunny	3	2
overcast	4	0
rainy	2	3

$P(x|C_i) = P(sunny|yes)$

$P(sunny|no)$

Likelihood table	Yes	No	
sunny	3/9	2/5	5/14
overcast	4/9	0/5	4/14
rainy	2/9	3/5	5/14
	9/14	5/14	

$P(X) = P(sunny)$

$P(X) = P(overcast)$

$P(X) = P(rainy)$

$P(C_i) = P(yes)$

$P(C_i) = P(no)$

Outlook	Play golf
rainy	no
rainy	no
overcast	yes
sunny	yes
sunny	yes
sunny	no
overcast	yes
rainy	no
rainy	yes
sunny	yes
rainy	yes
overcast	yes
overcast	yes
sunny	no

Class:

- play golf=yes: 9
- play golf=no: 5

$P(x|C_i) = P(sunny|yes)$ $P(sunny|no)$

Likelihood table	Yes	No		
sunny	3/9	2/5	5/14	$P(X) = P(sunny)$
overcast	4/9	0/5	4/14	$P(X) = P(overcast)$
rainy	2/9	3/5	5/14	$P(X) = P(rainy)$
	9/14	5/14		

$P(C_i) = P(yes)$ $P(C_i) = P(no)$

$$P(C_i|X) = \frac{P(X|C_i)P(C_i)}{P(X)}$$

$$P(yes|sunny) = \frac{\frac{3}{9} \times \frac{9}{14}}{\frac{5}{14}} = 0.6 \qquad P(no|sunny) = \frac{\frac{2}{5} \times \frac{5}{14}}{\frac{5}{14}} = 0.4$$

0.6 > 0.4: play golf=yes

Outlook	Yes	No	
sunny	3/9	2/5	5/14
overcast	4/9	0/5	4/14
rainy	2/9	3/5	5/14
	9/14	5/14	

Temperature	Yes	No	
hot	2/9	2/5	4/14
mild	4/9	2/5	6/14
cool	3/9	1/5	4/14
	9/14	5/14	

Humidity	Yes	No	
high	3/9	4/5	7/14
normal	6/9	1/5	7/14
	9/14	5/14	

Windy	Yes	No	
true	3/9	3/5	6/14
false	6/9	2/5	8/14
	9/14	5/14	

Outlook	Temperature	Humidity	Windy
rainy	cool	high	true

Class:
- play golf=yes: 9
- play golf=no: 5

- $X = (Outlook = rainy\ , Temperature = cool, Humidity = high, Windy = true)$
- $P(Yes|(Outlook = rainy\ , Temperature = cool, Humidity = high, Windy = true))$

$$= \frac{P(rainy|yes)P(cool|yes)P(high|yes)P(true|yes) \times P(yes)}{P(rainy)P(cool)P(high)P(true)}$$

- $P(no|(Outlook = rainy\ , Temperature = cool, Humidity = high, Windy = true)$

$$= \frac{P(rainy|no)P(cool|no)P(high|no)P(true|no) \times P(no)}{P(rainy)P(cool)P(high)P(true)}$$

Outlook	Yes	No	
sunny	3/9	2/5	5/14
overcast	4/9	0/5	4/14
rainy	2/9	3/5	5/14
	9/14	5/14	

Temperature	Yes	No	
hot	2/9	2/5	4/14
mild	4/9	2/5	6/14
cool	3/9	1/5	4/14
	9/14	5/14	

Humidity	Yes	No	
high	3/9	4/5	7/14
normal	6/9	1/5	7/14
	9/14	5/14	

Windy	Yes	No	
true	3/9	3/5	6/14
false	6/9	2/5	8/14
	9/14	5/14	

Outlook	Temperature	Humidity	Windy
rainy	cool	high	true

Class:
- play golf=yes: 9
- play golf=no: 5

- $P(rainy|yes)P(cool|yes)P(high|yes)P(true|yes) \times P(yes)$
$$= \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{9}{14} = 0.00529$$
- $P(rainy|no)P(cool|no)P(high|no)P(true|no) \times P(no)$
$$= \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{14} = 0.02057$$

$0.00529 < 0.02057$:
play golf? No

Avoiding the Zero-Probability Problem

- If $P(x_k|\mathrm{C}_i) = 0$ for any feature, the posterior score for class C_i becomes zero.
  - Consequently, class C_i cannot be predicted for that tuple.
- Laplacian correction (add-one smoothing)
  - Add 1 to the count of each possible feature value.

Optional: Handling Numeric Features

- For a continuous-valued feature x_k, $p(x_k \mid C_i)$ is usually computed based on Gaussian distribution with a mean μ and standard deviation σ (estimated from the training tuples)

$$p(x_k|C_i) = g(x_k, \mu_{C_i}, \sigma_{C_i}) = \frac{1}{\sqrt{2\pi}\sigma_{C_i}} e^{-\frac{(x_k-\mu_{C_i})^2}{2\sigma_{C_i}^2}}$$

  - x=35 (age), Class: buy_computer=yes, μ=38, σ=12:

$$P(age = 35|buycomputer = yes) = \frac{1}{\sqrt{2\pi}12} e^{-\frac{(35-38)^2}{2\times 12^2}}$$